

RYAN BLACKWOOD

Industrial Controls • Instrumentation • Technical Systems • Manufacturing Technology

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Portfolio: Networking • Automation • CAD • Manufacturing Systems

PROFESSIONAL SUMMARY

Technical professional with hands-on experience supporting Factory Acceptance Testing (FAT), industrial fire and gas detection systems, instrumentation validation, manufacturing operations, and technical infrastructure projects. Experience with Eagle Quantum Premier (EQP), 4–20 mA instrumentation loops, HART communications, Emerson TREX communicator, LonWorks networks, electrical troubleshooting, device wiring, system validation, CAD design, additive manufacturing, virtualization, and enterprise networking.

CORE COMPETENCIES

Factory Acceptance Testing (FAT) • Fire & Gas Detection • Eagle Quantum Premier (EQP) • 4–20 mA Instrumentation • HART Protocol • Emerson TREX • LonWorks • Device Wiring • Electrical Troubleshooting • Fault Simulation • Multimeters • Laboratory Power Supplies • Proxmox • OPNsense • VLANs • Routing • VPNs • FreeCAD • Resin 3D Printing

TECHNICAL EXPERIENCE

Detector Electronics — Material Handler | Oct 2024 – Present

Support engineering and manufacturing teams producing industrial safety systems. Assisted a Test Engineer for approximately five months with FAT and system validation activities. Built and configured test benches for troubleshooting and verification. Tested 4–20 mA instrumentation loops and HART-enabled devices using an Emerson TREX communicator. Utilized multimeters and laboratory power supplies, simulated electrical faults, validated EQP systems, gas detectors, flame detectors, smoke detectors, horns, strobes, and field devices. Performed detector verification using test films, calibration magnets, and alignment tools. Worked with LonWorks communications and industrial field devices while collaborating with engineering, testing, and production teams.

PROFESSIONAL EXPERIENCE

Best Buy — Warehouse Team Lead | Aug 2021 – Apr 2023

Led inventory, logistics, workflow execution, and operational improvements while maintaining productivity and accuracy.

Geek Squad — Repair Agent | Feb 2020 – Aug 2021

Diagnosed and repaired computer hardware and software systems, performed troubleshooting, and validated functionality.

Geek Squad — Consultation Agent | Jun 2018 – Feb 2020

Provided technical troubleshooting, diagnostics, and customer support.

TECHNICAL PROJECTS

Enterprise Homelab Infrastructure: Designed and deployed Proxmox virtualization and OPNsense firewall infrastructure with VLAN segmentation, VPN connectivity, routing, DNS services, and hosted applications.

CAD & Additive Manufacturing: Designed functional components using FreeCAD and produced prototypes using resin-based additive manufacturing workflows.

Manufacturing Systems & Automation: Developed concepts for barcode inventory systems, manufacturing dashboards, workflow optimization, and automation projects.

EDUCATION

Pinellas Gulf Coast Academy — High School Diploma (2020)

PROFESSIONAL DEVELOPMENT

Actively pursuing OSHA 10 General Industry, NFPA 70E, Industry 4.0 training, networking, and industrial automation education.

TECHNICAL PORTFOLIO

Scan the QR code to view enterprise networking projects, Proxmox and OPNsense infrastructure, CAD designs, additive manufacturing work, automation concepts, manufacturing systems projects, and technical documentation.